

Stochastic control

eg. Merton portfolio application

1 stock with BS dynamics

$$ds_t = \mu s_t dt + \sigma s_t dW_t$$

non-risky asset with const r

$$ds^0_t = r s^0_t dt$$

Hedging portfolio

θ_t = amount invested in stock at time t

V_t = value of port

$V_t - \theta_t$ = money in bank

self-financing condition (PZL)

$$dV_t = \underbrace{\frac{\partial V}{\partial s_t}}_{\text{shares at } t} ds_t + (V_t - \theta_t) r dt$$

Hedging portfolio θ^* st. $V_T = g(s_T)$

V_0 = price of g (non-arb)

Or $v(g) := \inf \{ V_0 : V_T \geq g(s_T) \text{ a.s. for } \theta \}$

$\pi(g)$ = market price of $g(s_T)$

$$-V(-g) \leq \pi(g) \leq V(g)$$

$V(g)$ is linear $-V(-g) = V(g)$

$$\therefore \pi(y) = v(y)$$

$$dv_t = (rv_t + \underbrace{\theta_t(\mu - r)}_{\text{action of the controller}})dt + \theta_t \sigma dw_t$$

action of the controller

$$\tilde{v}_t = e^{-rt} v_t, \quad \tilde{s}_t = e^{-rt} s_t, \quad \tilde{\theta}_t = e^{-rt} \theta_t$$

$$d\tilde{v}_t = \tilde{\theta}_t \frac{d\tilde{s}_t}{\tilde{s}_t}$$

$$\tilde{v}_T = v_0 + \int_0^T \tilde{\theta}_t \frac{d\tilde{s}_t}{\tilde{s}_t}$$

Optimization $\rightarrow$ Algo

$$v(y) := \inf v_0 \quad v_0 + \int_0^T \tilde{\theta}_t \frac{d\tilde{s}_t}{\tilde{s}_t} \geq \tilde{g}(\tilde{s}_T)$$

Merton Port

$$\text{Given } v_0, \quad \max_{\theta} E V_T - \lambda \text{Var}(V_T)$$

$$E U(V_T)$$

$$U: \mathbb{R} \rightarrow \mathbb{R} \nearrow \curvearrowright$$

concave or S-shape

$$\begin{aligned} E U(V_T) &= U(EV_T) + \underbrace{E(V_T - EV_T)}_0 U'(EV_T) \\ &\quad + \underbrace{\frac{1}{2} E(V_T - EV_T)^2}_{\text{Var}} \underbrace{U''(EV_T)}_{< 0 \text{ for concave}} + \dots \end{aligned}$$

Merton $\max_{\theta} E[U(V_T)]$

$$dV_t = [rV_t + \theta_t(\mu - r)] dt + \sigma \theta_t dW_t$$

$$U(x) = \frac{x^p}{p} \quad p < 1$$

CPII model : $\theta^* = \pi^* V_t$

different lending & borrowing

$$dV_t = \theta_t \frac{dS_t}{S_t} + (V_t - \theta_t) \uparrow L dt - (V_t - \theta_t) \uparrow F dt$$

with consumption

$$dV_t = [rV_t + \theta_t(\mu - r)] dt + \sigma \theta_t dW_t - c dt$$

$$\max_{\theta, c \geq 0} E e^{\rho T} U(V_T) + \int_0^T e^{\rho t} u(c_t) dt$$

$$\begin{aligned} \max E & \left[\int_0^T e^{-\int_0^s h(s, x_s, m_s) ds} \uparrow (c, x, m) ds \right. \\ & \left. + e^{-\int_0^T k(s, x_s, m_s) ds} g(x_T) \right] \end{aligned}$$

control process

$$dx_s = b(s, x_s, m_s) ds + \sigma(s, x_s, m_s) dW_s$$

controlled state

Chap 1 : Main tools

$(\Omega, \mathcal{F}, P)$

$W : \mathbb{R}^+ \times \Omega \rightarrow \mathbb{R}$ is BM if

① $W(0) = 0$

② $W_t - W_s$ indep of $\mathcal{F}_s$

③ $W_t - W_s \sim N(0, t-s)$

$\rightarrow W$ has continuous paths a.s.

W is BM in $\mathbb{R}^d$ if $W = \begin{pmatrix} w^1 \\ \vdots \\ w^d \end{pmatrix} : w^i$ indep BM

• $\text{Cov}(W_t, W_s) = s \wedge t$

• $E[W_t | \mathcal{F}_s] = W_s$ (Martingale)

• $E[\phi(W_t) | \mathcal{F}_s] = E[\phi(W_t + W_s - W_s) | \mathcal{F}_s, W_s]$
Markovian $= E[\phi(W_t + W_s - W_s) | W_s]$

• Symmetry W is a BM and

AW is a BM for all A

s.t. $AA^T = I$ orthogonal mat

• $B_t = W_T - W_{T-t}, \quad t \leq T$

$B_t := \frac{1}{\sqrt{t}} W_{at}, \quad t \geq 0 \rightarrow \text{BM}$

$\begin{cases} B_t = t W \frac{1}{t} & t > 0 \\ B_0 = 0 \end{cases} \quad \text{BM}$

• $\limsup_{t \rightarrow \infty} W_t = +\infty$ a.s.

BM nowhere differentiable

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{W_{t+h} - W_t}{h} &= \lim_{h \rightarrow 0} \frac{W_h}{h} \quad t = \frac{1}{h} \\ &= \lim_{t \rightarrow \infty} \underbrace{t \frac{W}{t}}_{\text{BM}} \\ &= \uparrow \infty \text{ a.s.} \end{aligned}$$

quadratic variation

$$\pi^n := t_0 = 0 < t_1 < t_2 \dots < t_n$$

$$QV_t^{\pi^n}(W) := \sum_{i \geq 1} (W_{t_i \wedge t} - W_{t_{i-1} \wedge t})^2$$

$$\xrightarrow{n \rightarrow \infty} t \text{ in } L^2$$

$$E(QV_t^{\pi^n} - t)^2$$

Quadratic variation

Partition of $\mathbb{R}_+$,
 $\pi^n := t_0 = 0 < t_1 < \dots < t_n < \dots$

$|\pi^n| := \sup_{i \geq 1} (t_i - t_{i-1})$ dense: $\lim_{n \rightarrow \infty} |\pi^n| = 0$

Thm 1 $QV_t^{\pi^n}(W) := \sum_{i \geq 1} (W_{t_i \wedge t} - W_{t_{i-1} \wedge t})^2$

$\xrightarrow{n \rightarrow \infty} t \text{ in } L^2$

② W BM in $\mathbb{R}^d$

$QV_t^{\pi^n}(W) = \sum_{i \geq 1} (W_{t_i \wedge t} - W_{t_{i-1} \wedge t})(W_{t_i \wedge t} - W_{t_{i-1} \wedge t})^T$

$\xrightarrow{n \rightarrow \infty} t I_d \text{ in } L^2$

$E[(QV_t^{\pi^n} - t)^2]$

$= E\left[\sum_{i \geq 1} \underbrace{(W_{t_i \wedge t} - W_{t_{i-1} \wedge t})^2}_{=: \eta_i^2} - (t_{i-1} - t_{i-2})^2\right]$

$\quad \quad \quad \text{indep.} \quad \quad \quad E \eta_i = 0$

$= E \sum_{i \geq 1} \eta_i^2 + 2 \sum_{i < j} E \eta_i E \eta_j$

$= \sum_{i \geq 1} (t_{i-1} - t_{i-2})^2 E[d(0,1)^2 - 1]^2$

$= 2 \sum_{i \geq 1} (t_{i-1} - t_{i-2})^2 = E[d(0,1)^4] + 1 - 2d(0,1)^2 = 2$

$\leq 2|\pi^n| \sum_{i \geq 1} (t_{i-1} - t_{i-2}) = 2|\pi^n| t \rightarrow 0 \text{ as } n \rightarrow \infty$

$\limsup_{t \rightarrow \infty} W_t$

$[= \sup_t$

Nowhere

$\limsup_{t \rightarrow 0} \frac{W_t}{t}$

$\liminf_{t \rightarrow 0} \frac{W_t}{t}$

No formula $df(t, w_t) = \frac{\partial f}{\partial t}(t, w_t) dt + \frac{\partial f}{\partial w}(t, w_t) dw_t$

$$+ \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, w_t) dt$$

Ito formula

$$df(t, w_t) = \frac{\partial f}{\partial t}(t, w_t) dt + \frac{\partial f}{\partial x}(t, w_t) dw_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, w_t) dt$$

Stochastic integral

Simple process

$$\phi_t = \sum_{i \geq 1} \varphi_{t_{i-1}} \mathbb{1}_{(t_{i-1}, t_i]}(t)$$

for some $\varphi_{t_{i-1}} \in \mathcal{F}_{t_{i-1}}$ -meas. b.b. and (t_i)

$$I(\phi) := \sum_{i \geq 1} \boxed{\varphi_{t_{i-1}}} (W_{t_i} - W_{t_{i-1}}) \quad (\text{Riemann Sum})$$

LEFT POINT